

CSE 4820 – ARDUINO MICROCONTROLLER CONCEPTS



COURSE DETAILS



Course Code

CSE 4820



Course Title

Arduino Microcontroller Concepts



Credits

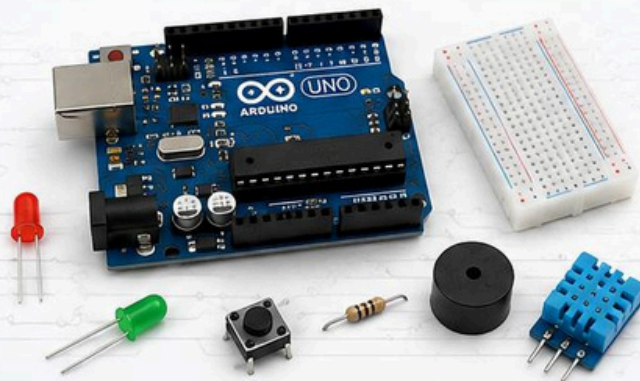
4



Weekly Contact Hours

3L – 2P

(3 Hours Theory + 2 Hours Laboratory)



MAJOR TOPICS COVERED

- Getting Started with Arduino
- Arduino Programming
- Mathematical Operations
- Serial Communication
- Digital Input and Output
- Analog Input and Output
- Sensor Interfacing
- Visual Output
- Physical Output Devices
- Audio Output
- Remote Control of External Devices
- Display Interfacing
- Time and Date Applications
- I2C Communication
- SPI Communication
- Simple Wireless Communication



SUGGESTED TEXTBOOK

O'REILLY®

Arduino Cookbook

Recipes to Begin, Expand, and Enhance Your Projects



Michael Margolis,
Nicholas Weldon & Brian Jepson

**Arduino Cookbook –
Recipes to Begin, Expand, and
Enhance Your Projects**

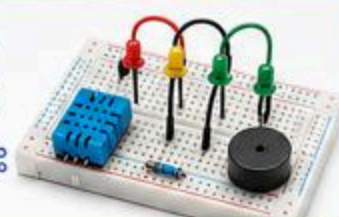
by Michael Margolis, Nicholas Weldon
& Brian Jepson

O'REILLY®



PROPOSED COURSE OUTCOMES

— CSE 4820 – Arduino Microcontroller Concepts —



CO NO.	COURSE OUTCOME	BLOOM'S TAXONOMY (L – LEARNING DOMAIN)	SIMPSON'S TAXONOMY (PSYCHOMOTOR DOMAIN)	KRATHWOHL'S TAXONOMY (AFFECTIVE DOMAIN)
C01	Explain the architecture, programming environment, and fundamental operation of Arduino microcontrollers, including digital and analog input/output concepts.	L2 – Understand	P1 – Perception	A1 – Receiving
C02	Develop Arduino programs using variables, mathematical operations, decision-making, loops, functions, and serial communication to solve basic engineering problems.	L3 – Apply	P3 – Guided Response	A2 – Responding
C03	Interface Arduino with sensors, switches, LEDs, buzzers, motors, and displays to acquire, process, and present real-time data.	L3 – Apply	P4 – Mechanism	A3 – Valuing
C04	Design and implement embedded systems using communication protocols such as I2C and SPI for integrating multiple peripheral devices.	L4 – Analyze	P5 – Complex Overt Response	A4 – Organization
C05	Analyze and troubleshoot Arduino hardware and software systems using debugging techniques, serial monitoring, and systematic testing methods.	L4 – Analyze	P6 – Adaptation	A4 – Organization
C06	Design and demonstrate a mini embedded application integrating sensors, actuators, displays, and communication modules while considering engineering ethics, sustainability, and societal needs.	L6 – Create	P7 – Origination	A5 – Characterization



BLOOM'S TAXONOMY (LEARNING DOMAIN)

L1 – Remember | L2 – Understand | L3 – Apply
L4 – Analyze | L5 – Evaluate | L6 – Create



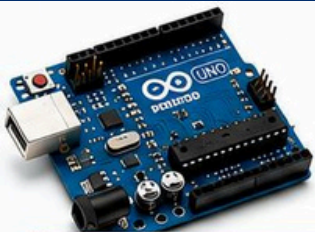
SIMPSON'S TAXONOMY (PSYCHOMOTOR DOMAIN)

P1 – Perception | P2 – Set | P3 – Guided Response
P4 – Mechanism | P5 – Complex Overt Response
P6 – Adaptation | P7 – Origination



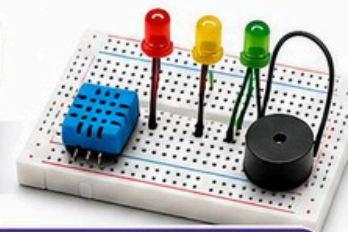
KRATHWOHL'S TAXONOMY (AFFECTIVE DOMAIN)

A1 – Receiving | A2 – Responding | A3 – Valuing
A4 – Organization | A5 – Characterization



TAXONOMY MAPPING – CSE 4820

ARDUINO MICROCONTROLLER CONCEPTS



BLOOM'S TAXONOMY (LEARNING DOMAIN)

LEVEL (L)	COURSE OUTCOMES (COs)
 L1 – Remember	–
 L2 – Understand	C01
 L3 – Apply	C02, C03
 L4 – Analyze	C04, C05
 L5 – Evaluate	–
 L6 – Create	C06



The course progresses from foundational understanding to analysis and creation of embedded systems.



SIMPSON'S TAXONOMY (PSYCHOMOTOR DOMAIN)

LEVEL (P)	COURSE OUTCOME (CO)
 P1 – Perception	C01
 P2 – Set	–
 P3 – Guided Response	C02
 P4 – Mechanism	C03
 P5 – Complex Overt Response	C04
 P6 – Adaptation	C05
 P7 – Origination	C06



Students advance from basic perception to origination, demonstrating high-level hands-on skills in real-world applications.



KRATHWOHL'S TAXONOMY (AFFECTIVE DOMAIN)

LEVEL (A)	COURSE OUTCOME (CO)
 A1 – Receiving	C01
 A2 – Responding	C02
 A3 – Valuing	C03
 A4 – Organization	C04, C05
 A5 – Characterization	C06



Learners move from awareness to internalizing values and developing a responsible and ethical approach to embedded systems.



CO KEY:

C01 – Explain fundamentals | C02 – Develop programs | C03 – Interface devices | C04 – Implement communication systems
C05 – Analyze & troubleshoot | C06 – Design & demonstrate embedded application



COURSE OUTCOME TO PROGRAM OUTCOME MAPPING

CSE 4820 – ARDUINO MICROCONTROLLER CONCEPTS



COURSE OUTCOMES (COs)		PROGRAM OUTCOMES (POs)										
		A	B	C	D	E	F	G	H	I	J	K
C01	Explain the architecture, programming environment, and fundamental operation of Arduino microcontrollers.	3	2	-	-	-	-	-	-	2	2	-
C02	Develop Arduino programs using variables, mathematical operations, and serial communication.	2	3	2	-	-	2	-	-	3	2	2
C03	Interface Arduino with sensors, actuators, displays to acquire, process, and present real-time data.	2	3	3	2	-	2	-	-	3	-	2
C04	Design and implement embedded systems using communication protocols like I2C and SPI.	2	3	3	2	-	2	-	-	3	3	3
C05	Analyze and troubleshoot Arduino hardware and software systems using debugging and testing methods.	2	3	2	-	2	2	-	2	3	2	2
C06	Design and demonstrate a mini embedded application considering ethics, sustainability, and societal needs.	2	2	3	3	3	3	2	3	3	2	3

CORRELATION SCALE

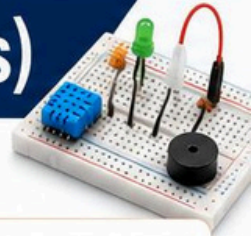
- 3 High Correlation
- 2 Moderate Correlation
- 1 Low Correlation
- No Direct Contribution

PROGRAM OUTCOMES (POs)

- | | |
|---|--|
| <p>A Apply knowledge of computing and mathematics</p> <p>B Analyze a problem and define computing requirements</p> <p>C Design, implement and evaluate computing solutions</p> <p>D Function effectively on teams</p> <p>E Understand professional, ethical, legal, security and social issues</p> <p>F Communicate effectively with a range of audiences</p> | <p>G Analyze local and global impact of computing</p> <p>H Engage in continuing professional development</p> <p>I Use current techniques, skills and tools necessary</p> <p>J Apply mathematical foundations and CS theory in design</p> <p>K Apply design and development principles in software systems</p> |
|---|--|



SUSTAINABLE DEVELOPMENT GOALS (SDGs)



— Aligned with CSE 4820 – Arduino Microcontroller Concepts —

4 QUALITY EDUCATION



SDG 4 – QUALITY EDUCATION

- Promotes experiential, hands-on learning through Arduino-based projects.
- Builds problem-solving, creativity, and technical skills for future engineers.



Relevance: Empowers students with practical knowledge and lifelong learning skills.

7 AFFORDABLE AND CLEAN ENERGY



SDG 7 – AFFORDABLE AND CLEAN ENERGY

- Enables development of energy-efficient systems and smart energy management solutions.
- Supports innovation in renewable energy monitoring and automation.



Relevance: Encourages sustainable energy use through smart embedded applications.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



SDG 9 – INDUSTRY, INNOVATION AND INFRASTRUCTURE

- Builds foundational skills for IoT, automation, and embedded system development.
- Encourages prototyping and innovation for real-world industrial applications.



Relevance: Drives innovation and supports resilient infrastructure through technology.

11 SUSTAINABLE CITIES AND COMMUNITIES



SDG 11 – SUSTAINABLE CITIES AND COMMUNITIES

- Supports smart city solutions like smart lighting, traffic monitoring, and environmental sensing.
- Helps create safer, smarter, and more sustainable urban spaces.



Relevance: Contributes to building inclusive, safe, resilient and sustainable communities.

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



SDG 12 – RESPONSIBLE CONSUMPTION AND PRODUCTION

- Encourages efficient use of resources and low-power design.
- Promotes reuse, recycling, and responsible electronic prototyping.



Relevance: Supports sustainable production and responsible use of technology.

13 CLIMATE ACTION



SDG 13 – CLIMATE ACTION

- Enables environmental monitoring (e.g., temperature, air quality, humidity).
- Supports data-driven solutions for climate change awareness and action.



Relevance: Empowers development of systems that monitor and protect our planet.



Overall Impact: This course nurtures responsible innovators who design technology-driven solutions that contribute to a sustainable and better future.



PERFORMANCE & ASSESSMENT PLAN

CSE 4820 – ARDUINO MICROCONTROLLER CONCEPTS



RECOMMENDED PERFORMANCE INDICATORS (PIs)

CO	PERFORMANCE INDICATOR	
C01		Explain Arduino hardware and software architecture and basic I/O concepts.
C02		Write, compile, and execute correct Arduino programs using control structures, functions, and serial communication.
C03		Interface sensors, actuators, displays, and output devices and acquire, process, and present real-time data.
C04		Develop embedded systems using I2C and SPI protocols to integrate multiple peripheral devices.
C05		Debug, test, and analyze Arduino hardware and software systems using systematic approaches.
C06		Design and demonstrate a mini embedded application that integrates sensors, actuators, displays, and communication modules to meet given requirements.



ASSESSMENT MAPPING

ASSESSMENT COMPONENT	COURSE OUTCOMES (COs)					
	C01	C02	C03	C04	C05	C06
 ATTENDANCE	✓	✓	✓	✓	✓	✓
 MAJOR LAB ASSIGNMENTS / QUIZZES	–	✓	✓	✓	✓	✓
 MINOR ASSIGNMENTS	✓	✓	✓	–	–	–
 MID-TERM	✓	✓	✓	✓	–	–
 IN-LAB EXAMINATION	–	–	✓	✓	✓	✓
 THEORY EXAMINATION	✓	✓	✓	✓	✓	–

ASSESSMENT COMPONENTS



Attendance



Major Lab
Assignments /
Quizzes



Minor
Assignments



Mid-Term



In-Lab
Examination



Theory
Examination

CORRELATION SCALE

- High Contribution
- Moderate Contribution
- Low Contribution
- No Direct Contribution



These performance indicators and assessments ensure students achieve the desired competencies aligned with the course outcomes.